

BARON LI

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PROJECTS

Traintopia | TypeScript, Svelte, Capacitor, WebGL, Fastify, Redis, Docker 2026 – Present

- Building a real-time NYC subway and bus tracking mobile app on a Capacitor/React frontend with a Fastify backend, delivering hyper-local live vehicle tracking beyond the native MTA and Google Maps transit experiences.
- Engineered a Redis- and BullMQ-backed job-queue pipeline to ingest, process, and serve live transit data feeds at scale.
- Developed an interactive 3D transit visualization with Three.js, architected to render 10,000+ simultaneous assets while targeting 30+ FPS.
- Designed a crowdsourced crowding-report feature using a 5-dot indicator backed by time-decay Bayesian aggregation.

Web Server & Streaming Platform | Python, Docker, MongoDB, WebSockets, FFmpeg Jan 2026 – May 2026

- Engineered a multi-threaded HTTP/1.1 web server from scratch with Python TCP sockets, building custom network parsers to process raw byte streams and multipart form data.
- Implemented the WebSocket protocol per RFC from the ground up, including the SHA-1 handshake and payload masking, to power a real-time multi-user drawing board.
- Built a secure authentication pipeline using bcrypt password hashing, SHA-256 session tokens, and TOTP-based two-factor authentication (2FA).
- Integrated FFmpeg into an adaptive bitrate streaming (ABR) pipeline, transcoding MP4 uploads into HTTP Live Streaming (HLS) formats with automatic thumbnail capture.

Encore - Karaoke Web Platform | React, Tailwind CSS, Howler.js, Web Audio AP Dec 2025

- Developed a responsive web app with real-time lyric synchronization and dual-track (instrumental/vocal) audio playback.
- Engineered a custom audio hook over Howler.js and the Web Audio API, manipulating GainNodes for independent vocal and instrumental volume mixing.
- Built a custom RegEx parser to convert LRC lyric files into an auto-scrolling teleprompter, using React refs for smooth DOM updates.
- Added a backend endpoint to fetch, process, and ingest YouTube links as playable karaoke tracks.

RecipeByte | HTML, CSS, JavaScript, PHP, MySQL, Apache, Git Aug 2025 – Dec 2025

- Built a dynamic, responsive full-stack web app for a software engineering course, letting users share and discover recipes in a community-driven platform.
- Engineered a secure user authentication system with registration, login, and a complete password-reset workflow in PHP and MySQL.
- Created an interactive recipe browsing page with search and filtering via JavaScript DOM manipulation.

Conway's Game of Life | C Aug 2024

- Implemented core cell-state transition logic for Conway's rules in C, emphasizing efficient multi-dimensional array handling and precise output formatting.

EXPERIENCE

Teaching Assistant | Police Athletic League, Queens, NY Jun 2021 – Aug 2021

- Supported classroom instruction for 30+ students, reviewing lessons, answering questions, and supervising during lunch, field trips, and recess.

EDUCATION

University At Buffalo - School of Engineering And Applied Science
Computer Science

Buffalo, NY
August 2023 – May 2027

TECHNICAL SKILLS

Languages: Java, C/C++, Python, OCaml, PHP, MySQL, JavaScript, HTML

Development/Frameworks: React, Tailwind CSS, Docker, MongoDB, WebSockets, Apache, Web Audio API

Tools: Git, VS Code, IntelliJ, Linux/Unix, FFmpeg, Nginx

Certifications: Google Data Analytics